

Week 1

Question 1

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n;
```

```
    int count = 2;
```

```
    if (scanf("%d", &n) != 1) {
```

```
        return 1;
```

```
    }
```

```
    int i = 1;
```

```
    int s = 1;
```

```
    while (count++, s <= n) {
```

```
        count++;
```

```

    i++;

    count++;

    s += i;
}

printf("%d\n", count);

return 0;
}

```

Output:

	Input	Expected	Got
	9	12	12
	4	9	9

Question 2

Convert the following algorithm into a program and find its time complexity using the counter method.

```

void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
void func(int n)
```

```
{
```

```
    int count = 0;
```

```
    count++;
```

```
    if(n == 1)
```

```
    {
```

```
        count++;
```

```
    }
```

```
    else
```

```
    {
```

```
        count++;
```

```
        for(int i = 1; i <= n; i++)
```

```
        {
```

```
            count++;
```

```
            for(int j = 1; j <= n; j++)
```

```
            {
```

```
                count++;
```

```
                count++;
```

```
                count++;
```

```
                break;
```

```
            }
```

```
            count++;
```

```
        }
```

```
    }
```

```
    printf("%d", count);
```

```
}
```

```
int main()
```

```

{
    int n;

    scanf("%d", &n);

    func(n);

    return 0;
}

```

Output:

Input	Expected	Got
2	12	12
1000	5002	5002
143	717	717

Question 3

Convert the following algorithm into a program and find its time complexity using counter method.

```

Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
}

```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
void Factor(int num)
```

```
{
```

```

int count = 0;
for(int i = 1; i <= num; ++i)
{
    count++;
    count++;
    if(num % i == 0)
    {
        count++;
    }
}
count++;
printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);
    Factor(n);
    return 0;
}

```

Output:

Input	Expected	Got
12	31	31
25	54	54
4	12	12

Question 4

Convert the following algorithm into a program and find its time complexity using counter method.

```

void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)

```

```

        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
    }

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
void function(int n)
```

```

{
    int c = 0;

    c++;

    for(int i = n / 2; (c++, i < n); i++)
    {
        for(int j = 1; (c++, j < n); j = 2 * j)
        {
            for(int k = 1; (c++, k < n); k = k * 2)
            {
                c++;
            }
        }
    }

    printf("%d", c);
}

```

```
int main()
```

```

{
    int n;

    scanf("%d", &n);

    function(n);

    return 0;
}

```

Output:

Input	Expected	Got
4	30	30
10	212	212

Question 5

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
void reverse(int n)
```

```
{
```

```
    int rev = 0, remainder;
```

```
    int count = 0;
```

```
    count++;
```

```
    while(n != 0)
```

```
    {
```

```
        count++;
```

```

    remainder = n % 10;

    count++;

    rev = rev * 10 + remainder;

    count++;

    n /= 10;

    count++;

}

count++;

count++;

printf("%d", count);

}

int main()

{

    int n;

    scanf("%d", &n);

    reverse(n);

    return 0;

}

```

Output:

Input	Expected	Got
12	11	11
1234	19	19